








Blockchain Applications

Lecture 5: Stablecoins II – Systemic Depegging Pressure and Portfolios



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Today's Question and Route

- **Question**: Lecture 4 watched one pool. Can we measure pressure on the *whole* stablecoin system in one number, and is that number useful for forecasting?
- **Route**
  - comparing tokens with different anchors: reference adjustment
  - quantile regression and the Lasso; the Financial Risk Meter (FRM) idea
  - the De-pegging Pressure Index (DPI): construction, regimes, events
  - inside the tail links: who is linked to whom
  - measure or forecast: two different tasks
- Main source: Fan, Sha, Härdle & Pele (2026), *Reference-adjusted Quantile Lasso reveals regime-dependent stablecoin depegging pressure*, draft; based on Fan (2026), MSc thesis, HU Berlin
- Own replication on End-of-Day Historical Data (EODHD), 10 stablecoins, 1 January 2020 – 23 October 2026

Learning Outcomes

- Compute a reference-adjusted peg deviation for dollar, euro and gold tokens
- Write down the check loss, the 5% Quantile Lasso and the generalised approximate cross-validation (GACV) criterion, and explain what the selected penalty measures
- Construct and interpret the DPI, its regimes and event responses, and judge their robustness
- Distinguish in-sample fit, out-of-sample forecasting and portfolio usefulness

Abbreviations

bp basis point, 0.01%
BUSD Binance USD (stablecoin)
CVIX crypto volatility index
DAI Dai (MakerDAO stablecoin)
DD drawdown (max DD: maximum drawdown)
df degrees of freedom
DPI De-pegging Pressure Index
DXY US dollar index
EODHD End-of-Day Historical Data (data provider)
ES expected shortfall
EUR euro
EURS STASIS Euro (stablecoin)
FABIZ Faculty of Business Administration in Foreign Languages, ASE
FRM Financial Risk Meter
FTX FTX crypto exchange (collapsed November 2022)
GACV generalised approximate cross-validation
GMV global minimum variance (portfolio)
GUSD Gemini Dollar (stablecoin)
HAC heteroskedasticity and autocorrelation consistent
HHI Herfindahl–Hirschman index
IDRT Rupiah Token (stablecoin)
L1 L1 norm, $\|\beta\|_1 = \sum_k |\beta_k|$ (Lasso penalty)
Lasso least absolute shrinkage and selection operator
MAD median absolute deviation
n.s. not significant
OOS out-of-sample
PAXG Pax Gold (gold-backed token)
pp percentage points
S&P 500 Standard & Poor's 500 stock index
SD standard deviation
SE standard error
sUSD Synthetix USD (stablecoin)
SVB Silicon Valley Bank (failed March 2023)
TUSD TrueUSD (stablecoin)
USD US dollar
USDC USD Coin (stablecoin)
USDP Pax Dollar (stablecoin)
USDT Tether (stablecoin)
VIX CBOE volatility index
XAU gold, one troy ounce

Reading and Tools

- Fan, Sha, Härdle & Pele (2026); Fan (2026), MSc thesis, HU Berlin, *Large Quantile Lasso FRM for Stable Coin Depegging Risks*; keynote *Stable Coin Risks* (Fan, Härdle, Pele, Sha, September 2026)
- FRM: Mihoci, Althof, Chen & Härdle (2020), Yu, Härdle, Borke & Benschop (2023); tail networks: Härdle, Wang & Yu (2016), Adrian & Brunnermeier (2016)
- Methods: Koenker & Bassett (1978) (quantile regression), Tibshirani (1996) (Lasso), Belloni & Chernozhukov (2011) (L1 quantile regression)
- Stablecoin arbitrage: Lyons & Viswanath-Natraj (2023)
- Video unit on Quantinar: FRM for Cryptos
- Estimation: Python (`scipy` linear programming, `joblib`); more than 48 000 Quantile Lasso equations, each on a grid of 25 penalties, estimated for this lecture

Reference Adjustment

- Not every stablecoin targets the dollar: EURS tracks the euro, PAXG one ounce of gold, IDRT the rupiah
- Let $P_{i,t}$ be the dollar price of token i and $B_{i,t}$ the dollar value of its reference asset:

$$d_{i,t} = 10^4 \left(\frac{P_{i,t}}{B_{i,t}} - 1 \right) \text{ bp}, \qquad r_{i,t} = 10^4 \, \Delta \ln \frac{P_{i,t}}{B_{i,t}}$$

  - $B = 1$ for dollar tokens; EUR/USD for EURS; XAU/USD for PAXG
- Worked example: EURS at 1.1194 USD while EUR/USD = 1.1202

$$d = 10^4 \left(\frac{1.1194}{1.1202} - 1 \right) = -7.1 \text{ bp}$$

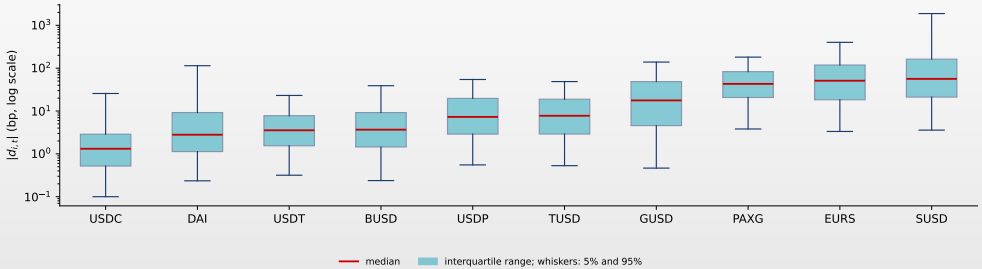
  - without adjustment, a 1% move of the euro would look like a 100 bp "depeg" of EURS
- The adjustment puts all tokens on one economic scale *before* any model sees them

Peg Quality, 2020 – May 2026

		own (EODHD)				own (EODHD)	
coin	med $\lvert d\rvert$	$q_{0.05}$	% $d \le -1\%$	paper med $\lvert d\rvert$	coin	med $\lvert d\rvert$	$q_{0.05}$	% $d \le -1\%$
USDC	1.3	−5.0	0.17	4.11	GUSD	17.7	−110.0	6.32
DAI	2.8	−9.0	0.68	7.06	PAXG	43.0	−124.0	9.22
USDT	3.6	−11.3	0.09	5.37	EURS	51.0	−251.9	17.06
BUSD	3.7	−8.6	0.09		sUSD	56.3	−1 883.4	25.99
USDP	7.3	−47.5	0.21	7.63				
TUSD	7.8	−37.8	2.09					

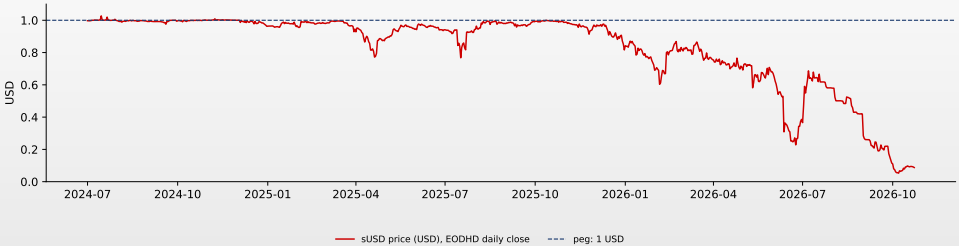
- Two tiers: large dollar tokens within a few bp; small, non-dollar or crypto-collateralised tokens tens of bp off, with deep lower tails
- The source matters for levels (EODHD medians are lower than CoinGecko's), not for the ranking; the downside rates agree closely (USDT 0.09% in both; sUSD 26.0% vs 25.0%)
- sUSD (crypto-collateralised): 0.95 USD in March 2025, 0.76 in March 2026, 0.09 on 23 October 2026, a persistent depeg

Two Tiers of Peg Quality (own)



- Distribution of $|d_{i,t}|$, January 2020 – May 2026, coins ordered by median; log scale
- The 95% whisker spans three orders of magnitude: 25 bp for USDC, almost 2 000 bp for sUSD

A Persistent Depeg: sUSD (own)



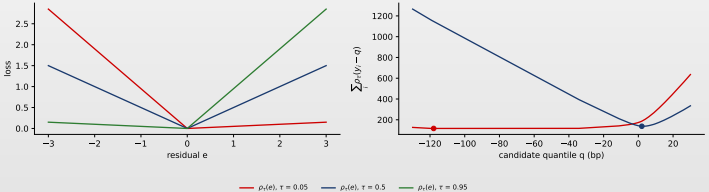
- Crypto-collateralised (Synthetix); below 0.99 USD on 607 of 661 days since January 2025; below 0.95 from 24 November 2025; 0.09 USD on 23 October 2026
- Unlike Silicon Valley Bank (SVB; two days), no arbitrage closes this gap: redemption at par is not available; a depeg can be a slow default

The Check Loss

- The τ-quantile minimises an asymmetric absolute loss (Koenker & Bassett (1978)):

$$\rho_\tau(e) = e\,\big(\tau - \mathbb{1}\{e < 0\}\big), \qquad q_\tau = \arg\min_q E\,\rho_\tau(Y - q)$$

- Quantile regression: $\min_{\alpha,\beta} \sum_i \rho_\tau(y_i - \alpha - x_i'\beta)$, a linear programme
- For $\tau = 0.05$ a residual below the line costs $0.95|e|$, above it $0.05|e|$: the fit is pushed to the lower tail



- Right: 20 daily deviations; for $\tau = 0.5$ the minimum is the median (2 bp); for $\tau = 0.05$ the loss is *flat* between -120 and -35 bp

Worked Example: How Many Points Define a 5% Quantile?

- With $n = 20$ and $\tau = 0.05$: $n\tau = 1$, so one observation lies below the quantile

$$\sum_i \rho_{0.05}(y_i - q) = 116.55 \quad \text{for every } q \in [-120, -35]$$

- Any value between the smallest and second-smallest observation is a valid 5% quantile: the estimate is not unique
- In a 90-day window $n\tau = 4.5$: four or five days carry the whole lower tail
- Interpretation
  - rolling 5% quantile models are noisy by construction; one day entering or leaving the window can move the estimate
  - this is why the DPI is analysed in regimes and event windows, not day by day

Quantile Lasso and the Selected Penalty

- With many candidate predictors, add an L1 penalty (Tibshirani (1996), Belloni & Chernozhukov (2011)):

$$(\hat{\alpha}, \hat{\beta})(\lambda) = \arg\min_{\alpha,\beta} \sum_{s \in W_t} \rho_{0.05}\left(d_{j,s} - \alpha - x'_{j,s}\beta\right) + \lambda \|\beta\|_1$$

- All slopes are zero for $\lambda \geq \lambda_{max}$, with $(\psi_\tau(u) = \tau - \mathbb{1}\{u < 0\})$

$$\lambda_{max} = \max_k \left| \sum_s x_{k,s}\, \psi_\tau\left(d_{j,s} - \hat{q}_\tau\right) \right|$$

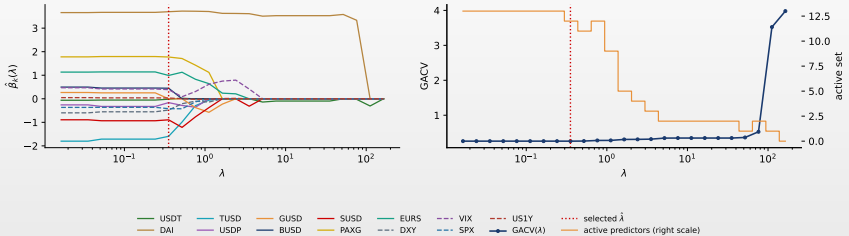
  - a scaled covariance between each predictor and the target's lower-tail indicator

- λ is chosen by generalised approximate cross-validation on a grid:

$$\mathrm{GACV}(\lambda) = \frac{\sum_s \rho_\tau\left(d_{j,s} - \hat{\alpha} - x'_{j,s}\hat{\beta}(\lambda)\right)}{n - \mathrm{df}(\lambda)}, \qquad \mathrm{df} = \#\{\hat{\beta}_k \neq 0\} + 1$$

- FRM idea (Mihoci, Althof, Chen & Härdle (2020)): the selected $\hat{\lambda}_{j,t}$ is a risk measure; higher $\hat{\lambda}$ = stronger tail co-movement under the fixed specification

Worked Example: One Equation, USDC after SVB (own)



- Target USDC, 90 days to 19 March 2023, 13 candidates (9 coins, 4 lagged macro changes), all scaled by the median absolute deviation (MAD)
- $\lambda_{max} = 164.1$; GACV minimum at $\hat{\lambda} = 0.353$ with 12 active predictors; largest slopes: DAI 3.70, PAXG 1.77, TUSD -1.59
- DAI enters first and stays to the largest λ: in the SVB window USDC's lower tail was DAI's lower tail

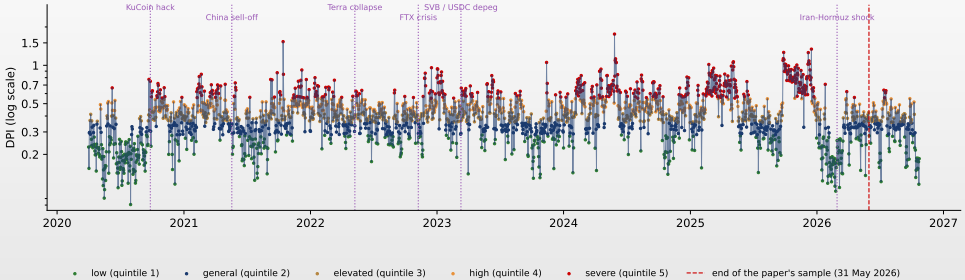
Construction

- For each coin j and day t, a 5% Quantile Lasso on the preceding 90 days
  - target: $d_{j,s}$; predictors: the other coins' $d_{k,s}$ (same day) and one-day-lagged changes of the US dollar index (DXY), the CBOE volatility index (VIX), S&P 500, 1-year Treasury yield (paper also: CVIX)
  - every variable centred by its window median and divided by $1.4826 \times \mathrm{MAD}$
  - 25 values of λ; $\hat{\lambda}_{j,t}$ minimises GACV
- Index:

$$\mathrm{DPI}_t = \frac{1}{J} \sum_{j=1}^{J} \hat{\lambda}_{j,t}$$

- Paper: 11 coins (CoinGecko), 2 252 days to 31 May 2026. Own: 10 coins (IDRT has no EODHD data after 2023), 2 397 days to 23 October 2026, 23 970 equations
- The DPI is *not* a depeg probability or a percentage; it is a model-implied measure of lower-tail dependence

The DPI, April 2020 – October 2026 (own)



- Annual medians: 0.28 (2020), 0.38–0.40 (2021–2023), 0.46 (2024), 0.53 (2025), 0.33 (2026); highest regime in late 2025, coinciding with the start of the sUSD depeg
- Log-linear trend +0.051 per year (heteroskedasticity and autocorrelation consistent, HAC(90), standard error (SE) 0.034, $p = 0.14$): regimes, no secular trend, as in the paper (-0.004, $p = 0.78$)

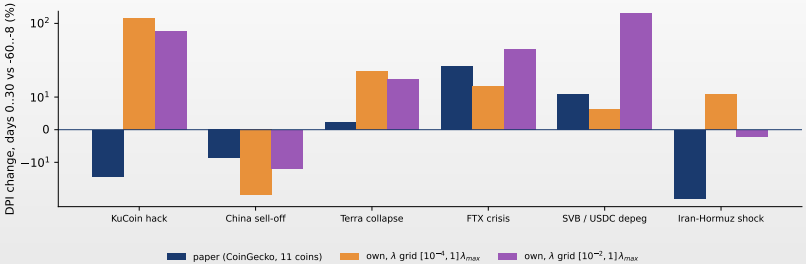
Event Responses

- Response: $\Delta = \overline{\mathrm{DPI}}_{[0,30]} / \overline{\mathrm{DPI}}_{[-60,-8]} - 1$; test on $\ln \mathrm{DPI}$ with HAC errors; Holm correction for six events

Event	date	paper	own	own Holm p	own, narrow λ grid
KuCoin hack	26 Sep 2020	−14.4%	+120.6%	< 0.001	+74.2%
China sell-off	19 May 2021	−8.6%	−20.2%	0.033	−11.9%
Terra collapse	9 May 2022	+2.2% (n.s.)	+17.9%	0.021	+15.4%
FTX crisis	8 Nov 2022	+19.5%	+13.4%	1.000	+36.7%
SVB / USDC depeg	11 Mar 2023	+11.0%	+6.1%	1.000	+152.1%
Iran–Hormuz shock	28 Feb 2026	−22.9%	+10.9%	1.000	−2.1%

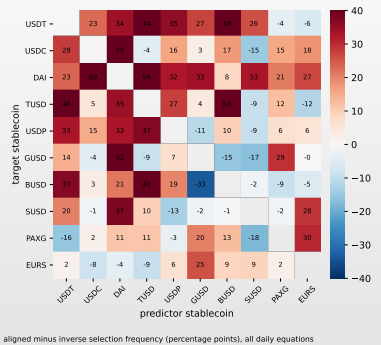
- Paper: significant rises only after FTX and SVB, the two events about intermediation and reserve confidence
- Own replication: the sign agrees for 4 of 6 events, but sizes and significance change with the data source and even with the λ grid

How Robust Are Event Responses?



- Narrow grid $[10^{-\mathbf{2}}, 1]\lambda_{max}$: 19.1% of equations select the smallest λ (a boundary solution); wide grid $[10^{-\mathbf{4}}, 1]\lambda_{max}$: 3.8%
- The two indices correlate 0.83 in logs, yet the SVB response is $+6\%$ or $+152\%$: a 31-day mean of a noisy index is sensitive to tuning choices
- Lesson: report event responses with their specification; trust the results that survive several specifications

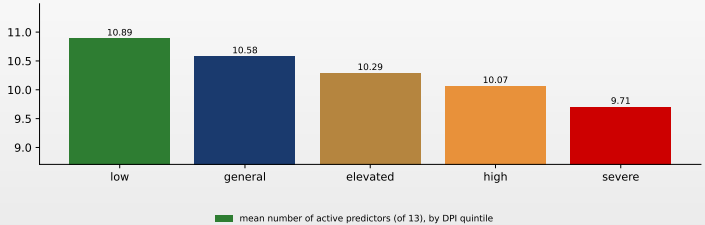
Who Is Linked to Whom? (own)



aligned minus inverse selection frequency (percentage points), all daily equations

- Positive slope = *aligned*: a lower predictor deviation goes with a lower target quantile
- Aligned 47.2% vs inverse 33.2% of coin coefficients (paper 39.6% vs 23.9%)
- Strongest pairs: USDC ← DAI (75 pp), TUSD ← BUSD (63), DAI ← USDC (61)
- Most inverse: BUSD ← GUSD (−34)
- Macro variables: 18.5% of absolute slope weight (paper 20.8%)
- Paper: no pair difference survives multiple testing; rankings change between halves: common structure, no fixed sender

Sparser Models under Pressure



- Own: from 10.89 active predictors (of 13) in the lowest DPI quintile to 9.71 in the highest; paper: 10.11 to 9.00 (of 15)
- A higher penalty removes more predictors: under pressure, a few strong tail links dominate the equation
- Same direction in both data sets: one of the results that survives replication

Fit Is Not Forecast (Paper)

Model	conditional skill	same-day OOS skill	fully lagged OOS skill
macro only	8.3%	−12.5%	−12.5%
coins only	33.7%	−15.9%	−34.4%
joint	37.0%	−19.8%	−38.6%

- Held-out check loss, relative to a rolling intercept-only model; negative = worse than the benchmark
- The best in-window description (joint, 37%) is the worst forecast once predictors are dated $t-1$ (−38.6%)
- The DPI alone has negative out-of-sample (OOS) R^2 at horizons of 10 to 150 days; the crypto FRM forecasts crypto volatility (R^2 0.17–0.29) but adding the DPI does not help
- On-chain variables (Curve, Uniswap) enter 91% of equations, yet do not reduce held-out loss significantly: frequent selection is not forecast value
- The DPI and crypto FRM correlate only 0.11: stablecoin pressure is not crypto stress

Concentration: Exposure, Not a Daily Alarm (Paper)

- Normalised Herfindahl–Hirschman index (HHI) of the 11-coin sample: 0.640 (2020), 0.291 (2022), 0.513 (2026); on 31 May 2026 USDT 69.4%, USDC 28.0%
- Regression of $\ln \mathrm{DPI}_t$ on lagged concentration, controls and year effects (HAC(90)):
  - $+1$ standard deviation (SD) of HHI: -1.52% DPI $[-4.24\%, 1.27\%]$; Top-2 share: -2.01% $[-4.93\%, 1.01\%]$; Holm $p = 0.38$
- Without year effects, higher concentration goes with higher DPI; with them the link disappears: both trend with the market regime
- Interpretation
  - concentration tells you *how much* of the system depends on two issuers: use it for scenarios and limits (Lecture 1: HHI)
  - it does not tell you *when* pressure builds

Measure or Forecast?

Task	Evidence	Verdict
Measure the current tail structure	aligned $>$ inverse links; regimes without trend	DPI useful, as one layer of monitoring
Forecast	negative OOS skill; lagged models worse; DPI adds nothing to FRM	no
Event attribution	FTX and SVB in the paper; sizes change with source and λ grid	fragile: report the specification

- A state measure is not a leading indicator: test each claim separately

Summary

- Reference adjustment puts dollar, euro and gold tokens on one scale; peg quality splits into two tiers (USDC 1.3 bp median vs EURS 51 bp, sUSD 56 bp)
- A 5% quantile in a 90-day window rests on 4–5 observations: rolling tail models are noisy by construction
- The DPI averages GACV-selected Quantile Lasso penalties; it moves in regimes with no secular trend
- Good fit is not forecast: held-out skill is negative, and worse with more information

Interpretation Exercises

1. PAXG trades at 3 998 USD while gold is 3 994 USD. Compute nothing: explain why its dollar return is not a depeg measure and what the reference adjustment changes.
2. In a 90-day window the 5% check loss is flat between two values. What does this imply for the stability of a rolling 5% Quantile Lasso?
3. The DPI rose by 152% after SVB in one specification and by 6% in another. Which results of the lecture would you still report with confidence, and why?
4. The HHI coefficient is -1.52% with interval $[-4.24\%, 1.27\%]$ after year effects. What is the conclusion, and what is concentration still useful for?

- Next: Lecture 6, networks, sentiment and asset classes

Appendix: Further Material

Optional material for self-study. It is **not examined**.

- Data Check: Nine Bad Prints (own)
- Coins or Macro? A Shapley Decomposition
- Four Long-Only Rules
- Results: Covariance Beats Penalties
- Value and Drawdown, 2020–2026 (own)
- What the Minimum-Variance Portfolio Holds (own)
- Worked Example: Why the Penalty Is a Bad Weight

Data Check: Nine Bad Prints (own)

- Rule (as in Lecture 1): a daily move beyond 1 000 bp that reverses by at least 90% the next day is a recording error; the close is set to missing and filled from the past

coin	date	close (USD)	log return (bp)
sUSD	6 Aug 2025	0.2743	−12 446
EURS	15 Jun 2025	0.8788	−2 720
sUSD	6 Oct 2026	0.0724	+2 633
EURS	23 Jun 2025	0.9362	−1 953
sUSD	14 Feb 2020	0.8372	−1 566
four more	2020–2026		1 000 to 1 500

- sUSD on 6 August 2025: low 0.21, close 0.27, next open 0.27, next close 0.98, on low volume
- Before cleaning, this single print gave the linear-lambda portfolio a 31% drawdown and more than doubled the equal-weight volatility (8.6% vs 3.9%): one bad day can decide a portfolio comparison

Coins or Macro? A Shapley Decomposition

- Conditional skill of a model m: $S_m = 1 - \overline{\mathrm{GACV}}_m / \overline{\mathrm{GACV}}_0$ (0 = intercept only)
- Two-block Shapley value of the coin block:

$$\phi_{coin} = \tfrac{1}{2}\left[(S_{coin} - S_0) + (S_{joint} - S_{macro})\right], \qquad \phi_{coin} + \phi_{macro} = S_{joint}$$

	paper		own (every 7th day)	
Model	GACV	skill	GACV	skill
intercept only	0.1808	–	0.1666	–
macro only	0.1658	8.3%	0.1563	6.2%
coins only	0.1199	33.7%	0.1022	38.7%
joint	0.1139	37.0%	0.0969	41.8%
coin share of joint skill ϕ_{coin}/S_{joint}		84.3%		88.8%

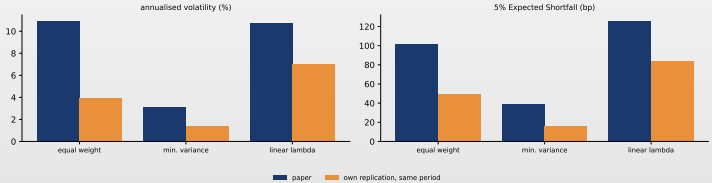
- Worked check (own): $\phi_{coin} = \tfrac{1}{2}[(38.7 - 0) + (41.8 - 6.2)] = 37.15$; $37.15/41.8 \approx 0.89$
- Robust result: the stablecoin system explains its own lower tail; macro variables add little

Four Long-Only Rules

- Holding reference-adjusted stablecoin positions (a treasury, a market maker, a lending desk); returns $r_{i,t}$ in bp
- Rules, rebalanced every 7 days with information to $t-1$, weights capped at 30%:
  - equal weight: $w_i = 1/J$
  - global minimum variance (GMV): $\min_w w'\hat{\Sigma}w$, $\sum w_i = 1$, $0 \le w_i \le 0.3$; $\hat{\Sigma}$ from the last 90 days
  - linear lambda: $\min_w \sum_i w_i \hat{\lambda}_{i,t-1}$ under the same constraints (fill the lowest-penalty coins first)
  - quadratic lambda (paper only): penalty level plus covariance
- Outcomes: annualised volatility, 5% ES of daily returns, maximum drawdown, turnover per rebalance

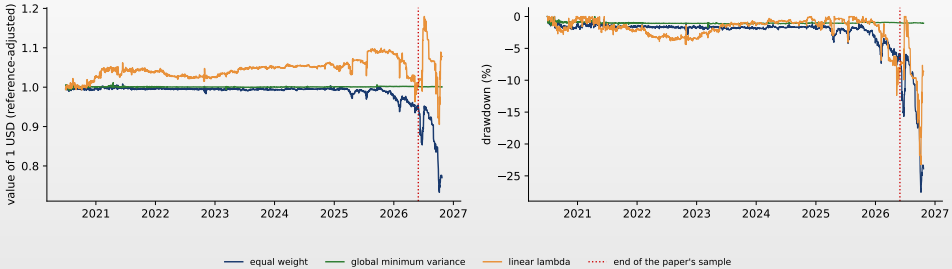
Results: Covariance Beats Penalties

	own, 30 Jun 2020 – 31 May 2026					paper		
Rule	return	vol	ES 5% (bp)	max DD	turnover	vol	ES 5%	max DD
equal weight	−0.81%	3.93%	49.4	7.6%	0.1%	10.93%	101.4	15.6%
min. variance	0.02%	1.44%	16.1	1.7%	3.7%	3.14%	39.0	2.6%
linear lambda	0.19%	7.06%	83.5	12.4%	59.2%	10.76%	125.7	16.5%



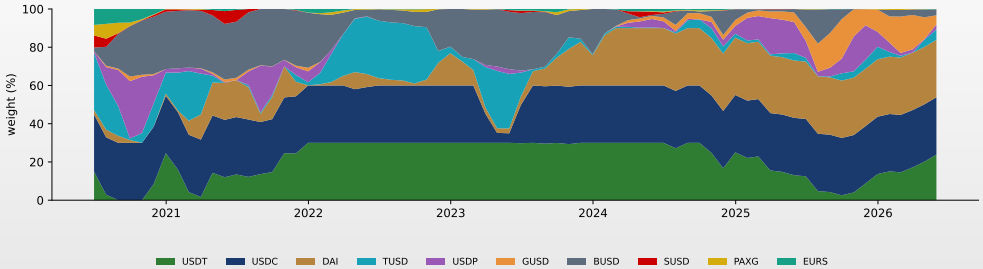
- Robust result: GMV cuts volatility by 63% and ES by 68% against equal weight (paper: 71%, 62%); penalty-based weights add risk and turnover

Value and Drawdown, 2020–2026 (own)



- GMV stays within 1.7% of its peak over six years; equal weight loses 27.6% once sUSD collapses (after the paper's sample)
- Linear lambda earns a little more in calm years and gives it back in stress: risk without a reliable reward

What the Minimum-Variance Portfolio Holds (own)



- Average weights: USDC 28.8%, USDT 21.1%, DAI 16.9%, BUSD 14.9%, TUSD 8.2%; sUSD, PAXG, EURS below 1% each
- The cap binds often: without it GMV would hold even more USDC and USDT, i.e. more issuer concentration

Worked Example: Why the Penalty Is a Bad Weight

- $\hat{\lambda}_{j,t}$ is the regularisation chosen for one equation; it depends on how the *other* coins move in j's lower tail, not on j's own risk
- A small, erratic token (high variance, few co-movements) can get a *low* $\hat{\lambda}$ and then a 30% weight under the linear rule
- In numbers (own): linear lambda volatility 7.1% vs 1.4% for GMV; turnover 59.2% per week: the ranking of $\hat{\lambda}$ changes almost every rebalance
- Extension to 23 October 2026 (sUSD falls from 0.39 to 0.09):

30 Jun 2020 – 23 Oct 2026	vol	ES 5% (bp)	max DD
equal weight	6.02%	76.9	27.6%
min. variance	1.39%	15.5	1.7%
linear lambda	10.66%	115.8	23.2%

- Equal weight more than triples its drawdown (7.6% to 27.6%) with sUSD's collapse; GMV had already moved out because sUSD's 90-day volatility exploded (364 bp a day before 12 June 2026)